Employee Burnout Analysis & Statistical Validation

Identifying Key Drivers of Workplace Burnout

The Growing Concern of Employee Burnout

To analyze workplace factors influencing burnout and provide actionable business recommendations.

- Employee burnout negatively impacts productivity and well-being.
- Organizations struggle to identify measurable drivers of burnout.
- Data-driven insights are needed to guide HR decisions.

Dataset Summary

Key Variables:

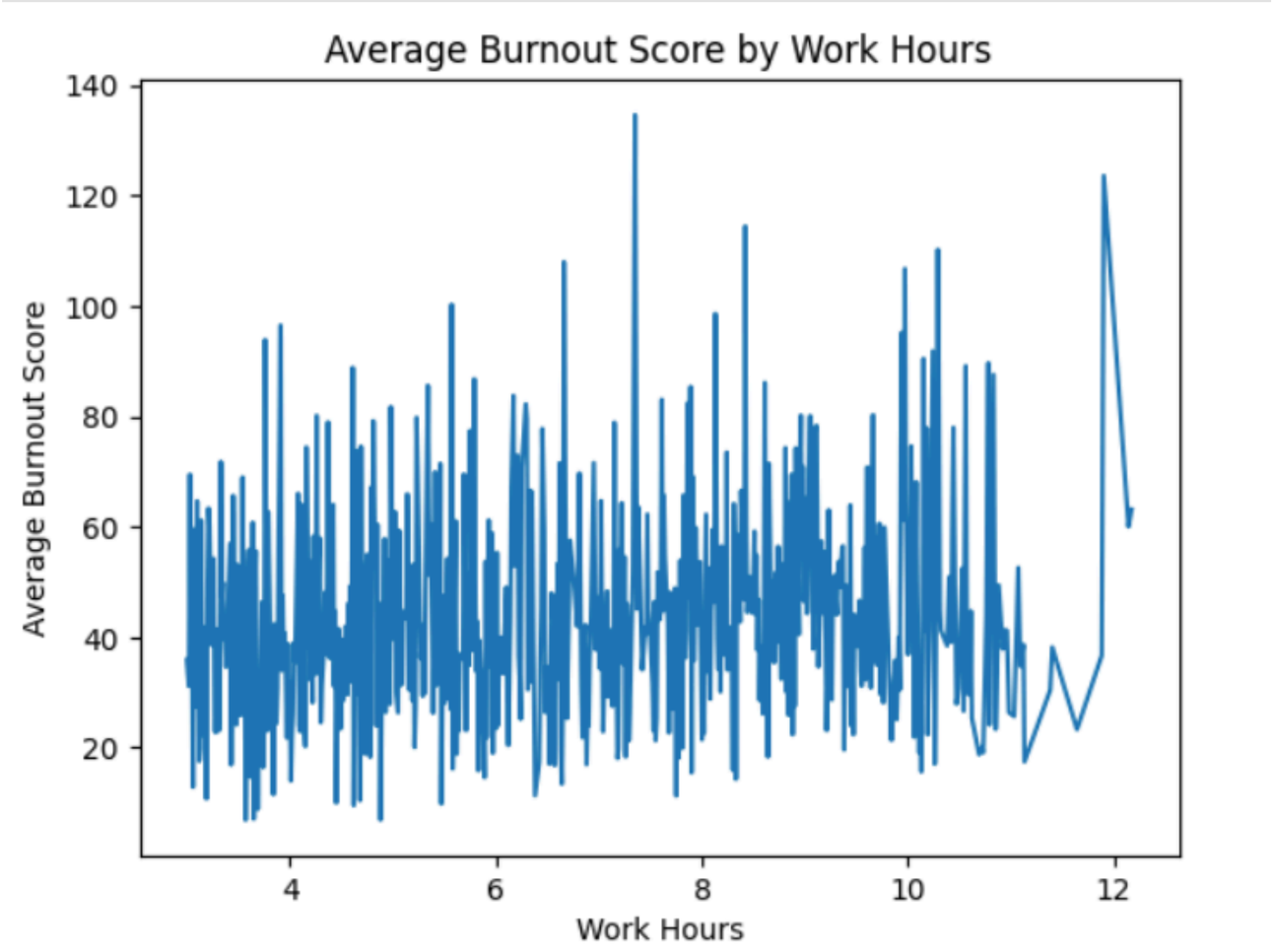
- Work Hours
- Screen Time
- Meetings Count
- Sleep Hours
- Work Intensity
- Burnout Score

Tools Used:

- Python
- Pandas
- Matplotlib/ Seaborn
- SciPy (Statistical Testing)

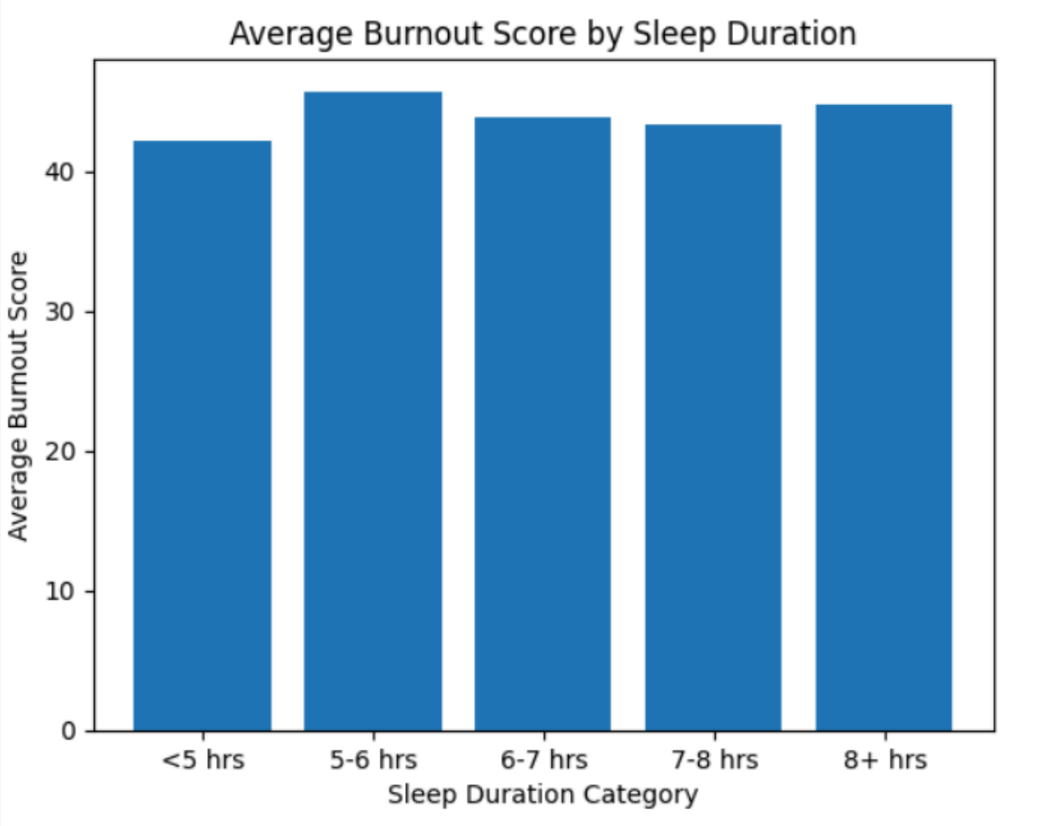
Work Hours & Burnout Trend

- Employees with longer work hours tend to show higher burnout scores.
- Positive relationship observed between work hours and burnout.
- Indicates workload plays a role in employee stress levels.



Sleep Hours & Burnout

- Lower sleep hours correlate with higher burnout levels.
- Employees with insufficient rest show increased stress indicators.
- Suggests importance of work-life balance.



Hypothesis Testing

Hypothesis

- H_0: Work hours do not significantly affect burnout score.
- H_1: Work hours significantly affect burnout score.

Test Used

- Pearson Correlation Test

Results

- Correlation Coefficient: 0.119
- p-value: 0.00000038

Conclusion

- Since p-value < 0.05, we reject the null hypothesis.
- Work hours have a statistically significant impact on burnout levels.

Business Interpretation

- Although the relationship is weak, it is statistically significant.
- Burnout is influenced by multiple factors, not just work hours.
- Extended working hours contribute to increased burnout risk.
- Organizations must monitor workload distribution carefully.

Actionable Recommendations

Monitor and regulate employee working hours.

Encourage structured breaks during workdays.

Promote flexible scheduling policies.

Implement employee wellness programs.

Track burnout indicators using internal analytics dashboards.

THANK YOU!